JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR
COLLEGE OF ENGINEERING (AUTONOMOUS)ANANTHAPURAMU– 515 002 (A.P) INDIA

B. Tech. – I Year I Semester
(Common to EEE, ECE, CSE)

Subject code	Title of the Subject	L	T	P	C
23A15302	Chemistry Lab	0	0	2	1

Course Objectives: Verify the fundamental concepts with experiments

Course Outcomes: At the end of the course, the students will be able to

- **CO1:** Determine the cell constant and conductance of solutions.
- **CO2:** Prepare advanced polymer Bakelite materials.
- **CO3:** Measure the strength of an acid present in secondary batteries.
- **CO4:** Analyze the IR spectra of some organic compounds.
- **CO5:** Calculate strength of acid in Pb-Acid battery.

List of Experiments:

1. Measurement of 10Dq by spectrophotometric method
2. Conductometric titration of strong acid vs. strong base
3. Conductometric titration of weak acid vs. strong base
4. Determination of cell constant and conductance of solutions
5. Potentiometry-determination of redox potential sandemfs
6. Determination of Strength of an acid in Pb-Acid battery
7. p^H metric titration of strong Acid Vs Strong Base
8. Preparation of a Bakelite
9. Verify Lambert-Beer's law
10. Wavelength measurement of sample through UV-Visible Spectro's copy
11. Identification of simple organic compounds by IR
12. Preparation of nano materials by precipitation method
13. Estimation of Ferrous Iron by Dichrometry

Learning outcomes:

At the end of the course, the students will be able to

- **Determine** the cell constant and conductance of solutions(L3)
- **Prepare** advanced polymer Bakelite materials(L2)
- **Measure** the strength of an acid present in secondary batteries(L3)
- **Analyze** the IR of some organic compounds(L3)

Reference:

- "Vogel'sQuantitativeChemicalAnalysis6thEdition6thEdition"Pearson Publications by J. Mendham, R. C. Denney, J. D. Barnes and B. Sivasankar